

notes, "Last year, eRocks was pretty good as a lot of open source companies had come. May be, this year it would be awesome." As a matter of fact, some really interesting presentations are expected this year (see box). Distributors are trying to support the community through open source development tools and boards being made available at an almost zero margin. "Kits'n'Spares would be selling the latest Raspberry Pi 'B+' boards at the lowest ever price in India," says the company's spokesperson Shyam Pandey. Codes to get discounts are being provided to design communities, for instance, through EFY's Electronics Design Community page at facebook.com/designelectronics

Semiconductor majors showing interest in open source. Semiconductor companies like Marvell, ATMEL, Broadcom, TI and even Intel are promoting open source to reach out to more designers who influence product management or decision making. In an article titled 'Intel CEO on Why He is Betting Big on the Maker Movement,' Brian Krzanich talks about open source communities and believes that there may be a 'next big thing' coming out of such communities. He says, "Those are the people who are inventing. Those are the creators. Those are the guys that are going to take something they see or do there and take it to Kickstart."

Plugcomputer (plugcomputer.org), Arduino, Raspberry Pi and Beagle series are some real examples of how semiconductor companies want to reach out to a larger designer base that can consider using their silicon as part of their designs. Moreover, the Internet of Things (IoT) has made it an even better avenue for such companies to come up with the designs along with add-on cards that enable connectivity and storage options.

Big industry stalwarts are 'opening' their products to ensure that, not just a group or team of engineers but an infinite audience can work on it and improve it. Earlier it was not

Interesting innovations from around the world

1. 3D printer that makes pizza for astronauts on long missions. Last year, Anjan Contractor, a mechanical engineer, was in the news for a US\$ 125,000 grant from NASA to build a prototype 3D printer. Now he has successfully built a prototype



Picture of the cooked pizza

2. Novena. Open source laptop. A new open hardware computing platform, flexible and powerful, designed for use as a desktop, laptop, or stand-alone board. Novena is a 1.2GHz, Freescale quad-core ARM architecture computer closely coupled with a Xilinx FPGA. It is designed for users who care about free software and open source, and/or want to modify and extend their hardware. All the documentation for the PCBs is open and free to download, the entire OS is buildable from source, and it comes with a variety of features that facilitate rapid prototyping.

For crafty people who want to build their case and define their own style, the main PCBA, stuffed with 4GB of RAM, 4GB microSD card and an Ath9k-based PCIe Wi-Fi card is available separately that boots to a Debian desktop over HDMI. Its fun features include:

- 100Mbit dual Ethernet capability allows laptop to be used as an in-line packet filter or router
- USB OTG enables laptop to spoof/fuzz Ethernet, serial, etc over USB via gadget interface to other USB hosts
- Utility serial EEPROM for storing crash logs and other bits of handy data
- Spartan-6 CGS324-packaged FPGA that has several interfaces to the CPU, including a 2Gbit/s (peak) RAM-like bus, for your bitcoin mining needs. Or whatever else you might want to toss in an FPGA.
- High-speed I/O expansion header useful for implementing a wide variety of functions, from simple GPIO breakouts to high-performance analogue data sampling front-ends

3. Spark Core: Arduino compatible Wi-Fi

development board. Spark Core is an Arduino-compatible, Wi-Fi enabled, cloud-powered development platform that makes creating Internet-connected hardware a breeze. This little board packs a punch with a 72MHz ARM Cortex M3, the best Wi-Fi module on the market, wireless programming, and lots of pin outs and peripherals. There's nothing you can't build with the Core.



3D printer in action



The board



Fully built laptop



Spark Core

What's new from vendors

1. Daughter cards/add-on boards compatible to known platforms.
2. Better support of software to enable focus on application development or problem solving – providing compatible / well-known IDE, debugger support and compatible add-on boards for rapid prototyping.
3. Industries are providing access to the design source details to bridge gap between time to market and R&D. However, the access is still limited, with terms and conditions.
4. Announcement by Parallax to open source its Propeller 1 P8X32A. This is a sign of things to come, and hopefully more vendors will follow. The Propeller 1 (P8X32A) is now a 100% open multi-core microcontroller, including all of the hardware and tools: Verilog code, Spin interpreter, PropellerIDE and SimpleIDE programming tools, and compilers. The Propeller 1 may be the most open chip in its class.